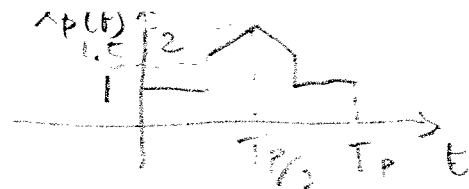
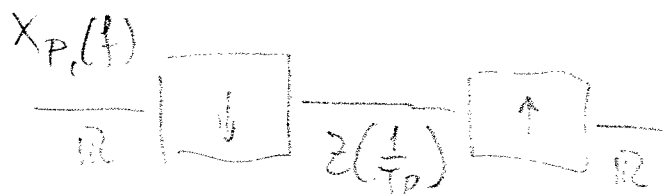
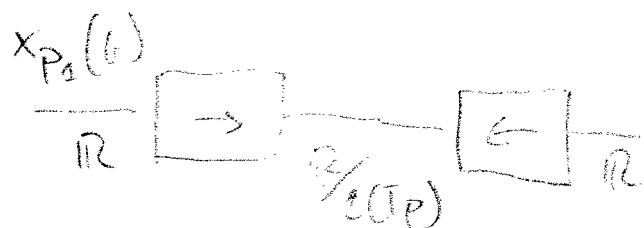


$$1) \quad x(t) = 1 + \sum_{k=-\infty}^{+\infty} x_{P_2}(t - kT_P)$$



$$x_{P_1}(t) = 0.5 \operatorname{rect}\left(\frac{t - T_P/2}{T_P/2}\right) + 0.5 \operatorname{triangle}\left(\frac{t - T_P/2}{T_P/4}\right)$$

$$X_{P_1}(f) = \frac{T_P}{4} \operatorname{sinc}\left(f \frac{T_P}{2}\right) e^{-j\pi f \frac{T_P}{2}} + \frac{T_P}{8} \operatorname{sinc}^2\left(f \frac{T_P}{4}\right) e^{-j\pi f \frac{T_P}{2}}$$



$$X(f) = \delta_{\mathbb{R}}(f) + \frac{1}{T_P} \sum_{k=-\infty}^{+\infty} X_{P_1}\left(\frac{k}{T_P}\right) \delta_{\mathbb{R}}\left(f - \frac{k}{T_P}\right)$$

$$X_{P_1}\left(\frac{k}{T_P}\right) = \left(\frac{T_P}{4} \operatorname{sinc}\left(\frac{k}{2}\right) + \frac{T_P}{8} \operatorname{sinc}^2\left(\frac{k}{4}\right) \right) (-1)^k$$

$$2) \quad x(kT) = x_1 * x_2(kT)$$

$$x_1(kT) = \frac{1}{T} 0.5^k 1_0(kT)$$

$$x_2(kT) = \begin{cases} 1 & k = -1, 0, \dots, 10 \\ 0 & \text{otherwise} \end{cases}$$

$$X(f) = X_1(f) X_2(f) \quad X_2(f) = 12T e^{-j\pi f \frac{11T}{2}} e^{j\pi f T}$$

$$X_1(f) = \frac{1}{T} \cdot \frac{T}{1 - 0.5 e^{-j\pi f T}} \operatorname{sinc}_{12}(12fT)$$

$$3) \quad e^{-\pi t^2} \leftrightarrow e^{-\pi f^2}$$

$$-j 2\pi t e^{-\pi t^2} \leftrightarrow -2\pi f e^{-\pi f^2}$$

$$-4\pi^2 t^2 e^{-\pi t^2} \leftrightarrow -2\pi (e^{-\pi f^2} - 2\pi f^2 e^{-\pi f^2})$$

$$t^2 e^{-\pi t^2} \leftrightarrow \frac{1}{2\pi} (e^{-\pi f^2} - 2\pi f^2 e^{-\pi f^2})$$

$$\text{area } t^2 e^{-\pi t^2} = \frac{1}{2\pi}$$

4) $x(t)$ è l'uscita di un filtro interpolatore con ingresso

$$x_1(kT) = 0.2^k 1_0(kT)$$

e risponde un polinomio $g(t) = \frac{1}{T}$ triangle $\frac{t}{T}$

$$x(t) = x_1(t) \cdot G(t) = \frac{1}{1 - 0.2 e^{-j 2\pi f T}} \sin c^2 \frac{f T}{1}$$

5)

$$H(f) = \frac{1+e}{2} \frac{1 - e^{-j 2\pi f T}}{1 - e^{-j 2\pi f T}} \quad e^{-j k T} = e^{j 2\pi k \frac{1}{2T}}$$

$$\begin{aligned} x(kT) &= 1 \cdot H(0) + H\left(\frac{1}{2T}\right) \cdot (-1)^k \\ &= 1 \cdot 0 + 1 \cdot (-1)^k = (-1)^k \end{aligned}$$

$$1) r_x(\tau) = R_0 B \sin c^2(\frac{f}{B})$$

$x(t) + x(t+t_0)$ è gaussiana e media nulla e varianza

$$E[x^2(t) + x^2(t+t_0) + 2x(t)x(t+t_0)] = 2r_x(0) + 2r_x(t_0) \\ = 2(R_0 B + R_0 B \sin c^2(\frac{1}{2})) \stackrel{\Delta}{=} \sigma^2$$

$$P[x(t) + x(t+t_0) < 2V_0] = \Phi\left(\frac{2V_0}{\sigma}\right)$$

$$2) c(t) \text{ è un coseno voltato con } (f) \text{ tale che} \\ C(0) = V_0 T \cdot A_H \Rightarrow c(0) = \text{area } C(f) = V_0 A_H = V_0'$$

$$\sigma_m^2 = R_0 E_h. \text{ Ora } |H(f)|^2 \text{ è un coseno voltato} \\ \text{con } |H(f)|^2|_{f=0} = T \Rightarrow E_h = 1 \Rightarrow \sigma_m = \sqrt{R_0}$$

$$P_e = \frac{4}{3} \cdot Q\left(\frac{V_0'}{2\sigma_m}\right)$$

3) Se $f_x(e)$ è esattamente costante all'interno di un intervallo di quantizzazione $\Rightarrow$

$$M_e = \frac{\Delta^2}{12} \quad \Delta = \frac{V}{2} \\ = \frac{V^2}{48}$$

$$h) F_c = 10 \text{ Msample/s} \quad \text{per } \text{SNR} > 45 \text{ dB} \Rightarrow$$

$$8 \text{ bit/sample} \quad \text{bit rate} = 80 \text{ Mbit/s.}$$

$$\text{bit/symbols PSK: } 2 \Rightarrow 40 \text{ Msymbols/s} \Rightarrow \text{bandwidth}$$

$$\text{maxime} = \frac{1}{T} = 40 \text{ MHz}$$

5)

$$r_y(\tau) = E[(s_u(t+\tau) + n_u(t+\tau))(s_u(t) + n_u(t))] =$$

$$= r_{s_u}(\tau) + r_{n_u}(\tau) \quad \text{in quanto}$$

$$E[n_u(t+\tau) s_u(t)] = E[s_u(t+\tau) n_u(t)] = 0$$

(indipendenza e medie nulle)

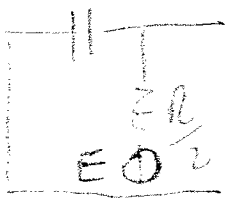
$$R_y(f) = R_{s_u}(f) + R_{n_u}(f)$$

Segnale

$$S_u = S \frac{R/2}{R/2 + \frac{1}{j2\pi f C}} = S \frac{j\pi f C R}{1 + j\pi f C R}$$

$$R_{s_u}(f) = R_s(f) \cdot \frac{\pi^2 C^2 R^2 f^2}{1 + \pi^2 f^2 R^2 C^2}$$

Rumore



$$N_u = E \frac{\frac{1}{j2\pi f C}}{\frac{R}{2} + \frac{1}{j2\pi f C}} = E \frac{1}{1 + j\pi f R C}$$

$$R_{n_u}(f) = 2kT_0 R \cdot \frac{1}{1 + \pi^2 f^2 R^2 C^2}$$